

DEPARTMENT OF MATHEMATICS & CONTROL
ENGINEERING

M.Sc. Applied Mathematics Program

Coursework Assignment 1: State-Space Modeling, Autonomous Dynamics & Linear Systems

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1 Task 1: Formulation of Discrete and Continuous State-Space Models

In this first task, I formulate mathematical state-space representations for discrete and continuous physical processes.

1.1 My Understanding of State Vectors

When I analyze a physical or economic system, I define its **state vector** $x(t) \in \mathbb{R}^n$ as the minimum set of variables required to fully describe the system's condition at time t . If I know $x(t)$ at the present moment, I can determine all future behavior using the state transition law without needing past history.

1.2 Physical Application: Damped Harmonic Oscillator

To demonstrate continuous-time modeling, I consider a mechanical particle moving under linear spring restoration and friction. Let $p(t)$ be its position and $v(t) = \dot{p}(t)$ be its velocity.

Position $p(t)$ alone is insufficient to predict future motion because the particle's direction of movement is unknown. Therefore, I construct the 2D state vector:

$$x(t) = \begin{bmatrix} p(t) \\ v(t) \end{bmatrix} \in \mathbb{R}^2$$

The second-order Newton law $\ddot{p} + \beta\dot{p} + k_0p = 0$ converts into my first-order matrix differential system:

$$\begin{bmatrix} \dot{p}(t) \\ \dot{v}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k_0 & -\beta \end{bmatrix} \begin{bmatrix} p(t) \\ v(t) \end{bmatrix}$$

Here, $A = \begin{bmatrix} 0 & 1 \\ -k_0 & -\beta \end{bmatrix}$ represents my system matrix.

1.3 Discrete-Time Financial Recurrence

For discrete-time dynamics, time progresses in integer steps $k \in \{0, 1, 2, \dots\}$. The evolution obeys a difference equation $x(k+1) = f(x(k))$.

My Numerical Example: I evaluate an investment account started with an initial capital of $x(0) = \$3,200$ earning an annual interest rate of 4.8%, compounded yearly.

$$x(k+1) = 1.048 x(k)$$

Iterating from my initial value $x_0 = 3200$:

$$x(1) = 1.048 \times 3200 = 3353.60$$

$$x(2) = 1.048 \times 3353.60 = (1.048)^2 \times 3200 = 3514.57$$

$$x(k) = (1.048)^k \cdot 3200$$

This closed formula allows me to compute the exact account balance at any year k .

2 Task 2: Comparative Analysis of Autonomous and Non-Autonomous Dynamics

In this task, I compare continuous dynamical systems based on explicit time dependence.

2.1 Autonomous Systems ($\dot{x} = f(x)$)

I classify a continuous system as **autonomous** when the vector field $f(x)$ depends solely on the state vector $x(t)$, with no explicit appearance of time t :

$$\frac{dx}{dt} = f(x)$$

Key Behavior I Observed:

1. **Time Invariance:** The vector field stays constant in time. Shifting my starting time from $t = 0$ to $t = t_0$ shifts the solution curve in time without altering its trajectory shape.
2. **Fixed Equilibria:** I determine equilibrium points x^* by solving $f(x^*) = 0$. At these points, $\dot{x} = 0$, so the system stays stationary.
3. **No Trajectory Crossings:** In a 2D phase portrait, solution curves never intersect one another due to existence-uniqueness properties.

2.2 Non-Autonomous Systems ($\dot{x} = f(x, t)$)

Conversely, a system is **non-autonomous** if time t appears explicitly in the derivative function:

$$\frac{dx}{dt} = f(x, t)$$

Key Behavior I Observed:

1. **Time-Varying Vector Fields:** The velocity field changes continuously over time, as seen in forced pendulums or seasonal biological models.
2. **Intersecting Spatial Projections:** 2D spatial projections of trajectories can cross because the vector field changes direction at identical points at different times.

Table 1: My Summary Table: Autonomous vs. Non-Autonomous Systems

Feature	Autonomous ($\dot{x} = f(x)$)	Non-Autonomous ($\dot{x} = f(x, t)$)
Explicit Time Variable	Absent	Present
Vector Field	Constant across time	Time-varying
Equilibrium Points	Fixed roots of $f(x^*) = 0$	Moving roots of $f(x^*, t) = 0$
Phase Trajectories	Non-crossing in 2D	Projections may intersect

3 Task 3: Linear State-Space Analysis and Exact Matrix Exponential Solutions

In this task, I investigate linear homogeneous systems $\dot{x}(t) = Ax(t)$ with initial condition $x(0) = x_0$.

3.1 Uncoupled 2D Linear System

When A is a diagonal matrix $A = \text{diag}(\lambda_1, \lambda_2)$, the system uncouples into independent scalar ODEs.

My Uncoupled Example: I consider the planar system:

$$\begin{cases} \dot{x}_1 = -2x_1 \\ \dot{x}_2 = 3x_2 \end{cases} \implies A = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}$$

Solving each equation independently gives:

$$x_1(t) = c_1 e^{-2t}, \quad x_2(t) = c_2 e^{3t}$$

To find the phase plane curves, I express e^t in terms of x_1 : $e^{-t} = (x_1/c_1)^{1/2} \implies e^t = (c_1/x_1)^{1/2}$. Substituting into x_2 :

$$x_2(t) = c_2 \left(\frac{c_1}{x_1(t)} \right)^{3/2} = \frac{C_0}{x_1^{1.5}}, \quad \text{where } C_0 = c_2 c_1^{1.5}$$

As $t \rightarrow \infty$, $x_1(t) \rightarrow 0$ and $x_2(t) \rightarrow \pm\infty$. The origin $(0, 0)$ is an **unstable saddle point**.

3.2 Matrix Exponential and Solution Uniqueness

For general square matrices $A \in \mathbb{R}^{n \times n}$, I define the matrix exponential via the power series:

$$e^{At} = \sum_{j=0}^{\infty} \frac{A^j t^j}{j!} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

My Uniqueness Proof Check: To prove that $x(t) = e^{At}x_0$ is the unique solution to $\dot{x} = Ax, x(0) = x_0$:

1. I differentiate e^{At} : $\frac{d}{dt}e^{At} = Ae^{At}$. Thus $x'(t) = Ae^{At}x_0 = Ax(t)$, so it satisfies the differential equation.
2. Let $z(t)$ be any other solution. I define $w(t) = e^{-At}z(t)$.
3. Differentiating $w(t)$: $w'(t) = -Ae^{-At}z(t) + e^{-At}z'(t) = -Ae^{-At}z(t) + e^{-At}Az(t) = 0$.
4. Since $w'(t) = 0$, $w(t)$ is constant. At $t = 0$, $w(0) = z(0) = x_0$. Therefore $w(t) = x_0 \implies z(t) = e^{At}x_0$. This confirms uniqueness!

3.3 My Fully Worked Numerical Problem

Problem Statement: Solve the coupled linear system:

$$\begin{cases} \dot{x}(t) = 3x(t) + 2y(t) \\ \dot{y}(t) = 4x(t) + 1y(t) \end{cases} \quad \text{with initial condition } x(0) = 4, y(0) = -2$$

Step 1: Characteristic Equation and Eigenvalues Writing in matrix form $\dot{X} = AX$ with $A = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$:

$$\det(A - \lambda I) = \det \begin{bmatrix} 3 - \lambda & 2 \\ 4 & 1 - \lambda \end{bmatrix} = (3 - \lambda)(1 - \lambda) - 8 = \lambda^2 - 4\lambda - 5 = 0$$

Factoring gives $(\lambda - 5)(\lambda + 1) = 0$, yielding eigenvalues:

$$\lambda_1 = 5, \quad \lambda_2 = -1$$

Step 2: Eigenvector Calculation

- For $\lambda_1 = 5$: $(A - 5I)v_1 = 0 \implies \begin{bmatrix} -2 & 2 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 0 \implies v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
- For $\lambda_2 = -1$: $(A + I)v_2 = 0 \implies \begin{bmatrix} 4 & 2 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = 0 \implies v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

Step 3: Modal Matrix and Inverse

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad \det(P) = -3 \implies P^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 4: State Solution $X(t) = Pe^{Dt}P^{-1}X_0$ First, compute transformed initial state $C_0 = P^{-1}X_0$:

$$C_0 = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 8 - 2 \\ 4 + 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Now multiply $X(t) = P \begin{bmatrix} 2e^{5t} \\ 2e^{-t} \end{bmatrix}$:

$$X(t) = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2e^{5t} \\ 2e^{-t} \end{bmatrix} = \begin{bmatrix} 2e^{5t} + 2e^{-t} \\ 2e^{5t} - 4e^{-t} \end{bmatrix}$$

My Final Exact Solution:

$$x(t) = 2e^{5t} + 2e^{-t}, \quad y(t) = 2e^{5t} - 4e^{-t}$$

Initial Condition Verification: At $t = 0$, $x(0) = 2 + 2 = 4$ and $y(0) = 2 - 4 = -2$. This matches my initial conditions perfectly!

4 Task 4: Annotated Course Reading Literature Review

In this task, I review key academic literature consulted during this coursework:

1. **Hirsch, M. W., Smale, S., & Devaney, R. L. (2012).** *Differential Equations, Dynamical Systems, and an Introduction to Chaos* (3rd ed.). Academic Press.

My Note: I consulted Chapters 1–3 for rigorous proofs of matrix exponentials and phase plane classifications.

2. **Perko, L. (2001).** *Differential Equations and Dynamical Systems* (3rd ed.). Springer-Verlag.

My Note: Used for state-space coordinate transformations and canonical Jordan forms.

3. **Khalil, H. K. (2002).** *Nonlinear Systems* (3rd ed.). Prentice Hall.

My Note: Reference for non-autonomous system stability and time-varying vector field properties.

4. **Strogatz, S. H. (2014).** *Nonlinear Dynamics and Chaos*. Westview Press.

My Note: Provides valuable physical intuition for phase portraits and mechanical oscillator state vectors.